

possibility that, at least in some contexts, accessibility regulation could involve the modulation of putative RSS-specific *trans*-acting factors that promote RAG-1/2 access. □

Methods

Generation of targeted ES cells and mutant mice

We generated pDJ β 2KO by subcloning the 4-kb *KpnI*–*AlwNI* 5' and 3-kb *XbaI*–*SpeI* 3' homology regions that flank the DJ β 2 gene cluster into the previously described pLNTK targeting vector²⁹. The construction of the targeting vectors for the M3, M4 and M5 mutations and the generation of ES cells with J β 1^{M3}, J β 1^{M4} and J β 1^{M5} mutant alleles are described in the Supplementary Information. Transfection and selection of ES cells were carried out as described²⁹. Chimaeric mice were generated by RAG-2-deficient blastocyst complementation¹⁹.

Southern and northern blot analyses

Southern and northern blot analyses were carried out as described²⁹. Probes A and C were 0.8-kb *DrdI*–*DrdI* and 0.7-kb *AflII*–*HaeII* TCR β genomic DNA fragments, respectively. The J β 1 probe was a 2-kb *PstI* genomic DNA fragment. Glyceraldehyde phosphate dehydrogenase and complementary CD3 ϵ DNA probes have been described previously²⁹.

PCR analyses

The V β primer set (PV) has been described previously³⁰ with the exception of the V β 14 primer: 5'-GGCAAGCAAGCTGGTGTGT-3'. The sequences of the remaining primers are as follows: P2, 5'-CCTGACTTCCACCCGAGGTT-3'; P3, 5'-CCTTCCTTATCTT-CAACTC-3'; P4, 5'-AAGGACGACTCTGTCTT-3'; PR2, 5'-CATCCTTCCTCTGAT-TAC-3'. PCR was carried out as described³⁰ at 92°, 90 s; 60°, 150 s; 72°, 60 s for 30 cycles.

Flow cytometric analysis and cell sorting

Flow cytometric analysis of thymocytes for CD4 and CD8 expression was carried out as described²⁹. Cell sorting was carried out by pretreatment of thymocytes with anti-CD4 and anti-CD8 and complement followed by sorting of CD25⁺/CD4⁺/CD8⁺ thymocytes on an Epics MoFlow sorter.

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1. Tonegawa, S. Somatic generation of antibody diversity. *Nature* **302**, 575–581 (1983).
2. Weaver, D. T. V(D)J recombination and double-strand break repair. *Adv. Immunol.* **58**, 29–85 (1995).
3. Jeggo, P. A. Identification of genes involved in repair of DNA double-strand breaks in mammalian cells. *Radiat. Res.* **150**, 580–591 (1998).
4. Eastman, Q. M., Leu, T. M. & Schatz, D. G. Initiation of V(D)J recombination *in vitro* obeying the 12/23 rule. *Nature* **380**, 85–88 (1996).
5. van Gent, D. C., Ramsden, D. A. & Gellert, M. The RAG1 and RAG2 proteins establish the 12/23 rule in V(D)J recombination. *Cell* **85**, 107–113 (1996).
6. Uematsu, Y. *et al.* In transgenic mice the introduced functional T cell receptor beta gene prevents expression of endogenous beta genes. *Cell* **52**, 831–841 (1988).
7. Ferrier, P. *et al.* Separate elements control DJ and VDJ rearrangement in a transgenic recombination substrate. *EMBO J.* **9**, 117–125 (1990).
8. Davis, M. M. & Bjorkman, P. J. T-cell antigen receptor genes and T-cell recognition. *Nature* **334**, 395–402 (1988).
9. Nadel, B., Tang, A., Escuro, G., Lugo, G. & Feeney, A. J. Sequence of the spacer in the recombination signal sequence affects V(D)J rearrangement frequency and correlates with nonrandom V κ usage *in vivo*. *J. Exp. Med.* **187**, 1495–1503 (1998).
10. Hesse, J. E., Lieber, M. R., Mizuuchi, K. & Gellert, M. V(D)J recombination: a functional definition of the joining signals. *Genes Dev.* **3**, 1053–1061 (1989).
11. Gerstein, R. M. & Lieber, M. R. Coding end sequence can markedly affect the initiation of V(D)J recombination. *Genes Dev.* **7**, 1459–1469 (1993).
12. Boubnov, N. V., Wills, Z. P. & Weaver, D. T. Coding sequence composition flanking either signal element alters V(D)J recombination efficiency. *Nucleic Acids Res.* **23**, 1060–1067 (1995).
13. Ramsden, D. A. & Wu, G. E. Mouse κ light-chain recombination signal sequences mediate recombination more frequently than do those of λ light chain. *Proc. Natl Acad. Sci. USA* **88**, 10721–10725 (1991).
14. Wei, Z. & Lieber, M. R. Lymphoid V(D)J recombination. Functional analysis of the spacer sequence within the recombination signal. *J. Biol. Chem.* **268**, 3180–3183 (1993).
15. Connor, A. M. *et al.* Mouse V β 1783 recombination signal sequences mediate recombination more frequently than those of V β 1558. *J. Immunol.* **155**, 5268–5272 (1995).
16. Akira, S., Okazaki, K. & Sakano, H. Two pairs of recombination signals are sufficient to cause immunoglobulin V-(D)-J joining. *Science* **238**, 1134–1138 (1987).
17. Gauss, G. H. & Lieber, M. R. The basis for the mechanistic bias for deletional over inversive V(D)J recombination. *Genes Dev.* **6**, 1553–1561 (1992).
18. VanDyk, L. F., Wise, T. W., Moore, B. B. & Meek, K. Immunoglobulin D β recombination signal sequence targeting: effect of D β coding and flanking regions and recombination partner. *J. Immunol.* **157**, 4005–4015 (1996).
19. Chen, J., Lansford, R., Stewart, V., Young, F. & Alt, F. W. RAG-2-deficient blastocyst complementation: an assay of gene function in lymphocyte development. *Proc. Natl Acad. Sci. USA* **90**, 4528–4532 (1993).
20. Willerford, D. M., Swat, W. & Alt, F. W. Developmental regulation of V(D)J recombination and lymphocyte differentiation. *Curr. Opin. Genet. Dev.* **6**, 603–609 (1996).
21. Sikes, M. L., Suarez, C. C. & Oltz, E. M. Regulation of V(D)J recombination by transcriptional promoters. *Mol. Cell. Biol.* **19**, 2773–2781 (1999).

22. Doty, R. T., Xia, D., Nguyen, S. P., Hathaway, T. R. & Willerford, D. M. Promoter element for transcription of unrearranged T-cell receptor beta-chain gene in pro-T cells. *Blood* **93**, 3017–3025 (1999).
23. Whitehurst, C. E., Chattopadhyay, S. & Chen, J. Control of V(D)J recombinational accessibility of the D β 1 gene segment at the TCR β locus by a germline promoter. *Immunity* **10**, 313–322 (1999).
24. Ramsden, D. A., Baetz, K. & Wu, G. E. Conservation of sequence in recombination signal sequence spacers. *Nucleic Acids Res.* **22**, 1785–1796 (1994).
25. Yancopoulos, G. D. *et al.* Preferential utilization of the most J μ -proximal V μ gene segments in pre-B-cell lines. *Nature* **311**, 727–733 (1984).
26. Jouvin-Marche, E. *et al.* Genomic organization of the T cell receptor V α family. *EMBO J.* **9**, 2141–2150 (1990).
27. Alt, F. W. *et al.* Ordered rearrangement of immunoglobulin heavy chain variable region segments. *EMBO J.* **3**, 1209–1219 (1984).
28. Rock, E. P., Sibbald, P. R., Davis, M. M. & Chien, Y.-H. CRD3 length in antigen-specific immune receptors. *J. Exp. Med.* **179**, 323–328 (1994).
29. Sleckman, B. P., Bardou, C. G., Ferrini, R., Davidson, L. & Alt, F. W. Function of the TCR alpha enhancer in $\alpha\beta$ and $\gamma\delta$ T cells. *Immunity* **7**, 505–515 (1997).
30. Gärtner, F. *et al.* Immature thymocytes employ distinct signaling pathways for allelic exclusion versus differentiation and expansion. *Immunity* **10**, 537–546 (1999).
31. Sleckman, B. P. *et al.* Mechanisms that direct ordered assembly of T-cell receptor β locus V, D, and J gene segments. *Proc. Natl Acad. Sci. USA*. (in the press).

Supplementary information is available on Nature's World-Wide Web site (<http://www.nature.com>) or as paper copy from the London editorial office of Nature.

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Intraprotein radical transfer during photoactivation of DNA photolyase

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Amino-acid radicals play key roles in many enzymatic reactions¹. Catalysis often involves transfer of a radical character within the protein, as in class I ribonucleotide reductase where radical transfer occurs over 35 Å, from a tyrosyl radical to a cysteine^{1–3}. It is currently debated whether this kind of long-range transfer occurs by electron transfer, followed by proton release to create a neutral radical, or by H-atom transfer, that is, simultaneous transfer of electrons and protons^{4–7}. The latter mechanism avoids the energetic cost of charge formation in the low dielectric protein^{4,5}, but it is less robust to structural changes than is electron transfer⁷. Available experimental data do not clearly discriminate between these proposals. We have studied the mechanism of photoactivation (light-induced reduction of the flavin adenine dinucleotide cofactor) of *Escherichia coli* DNA photolyase^{8–10} using time-resolved absorption spectroscopy. Here we show that the excited flavin adenine dinucleotide radical abstracts an electron from a nearby tryptophan in 30 ps. After subsequent electron transfer along a chain of three tryptophans,

the most remote tryptophan (as a cation radical) releases a proton to the solvent in about 300 ns, showing that electron transfer occurs before proton dissociation. A similar process may take place in photolyase-like blue-light receptors.

DNA photolyase catalyses the repair of major ultraviolet-induced DNA lesions, cyclobutane pyrimidine dimers or (6-4)-photoproducts, using the energy of near-ultraviolet light^{8,10} (Fig. 1). This enzyme, which is found in a variety of organisms ranging from bacteria to multicellular eukaryotes, consists of a single polypeptide with a relative molecular mass of ~55,000. It binds a flavin adenine dinucleotide (FAD) molecule as the essential catalytic cofactor, and it only functions when the FAD is fully reduced (FADH⁻). In isolated DNA photolyases, the FAD is typically in the inactive semireduced radical state FADH[•], but it can be easily reduced to the catalytically active FADH⁻ state by illumination with visible light in the presence of an exogenous electron donor, a process called photoactivation (Fig. 1).

The mechanism of photoactivation has been studied for *E. coli* photolyase^{9,11,12}. Excited FADH[•] oxidizes the solvent-accessible tryptophan W306, and exogenous electron donors then reduce the oxidized tryptophan (in competition with charge recombination occurring in about 10 ms; see Fig. 1, broken arrow) and thus stabilize the FADH⁻ state as required for the DNA repair function of the enzyme. We used absorption spectroscopy in the picosecond to millisecond time domain after selective excitation of FADH[•] by red laser flashes to study in detail the mechanism of radical transfer from FADH[•] to tryptophan (TrpH; H denotes the N1 proton) in recombinant *E. coli* photolyase¹³.

The final protonation state of the tryptophanyl radical was determined from absorption changes accompanying the recombination between FADH⁻ and the tryptophanyl radical in the absence of an exogenous electron donor (Fig. 2). Throughout the 400–645-nm spectral range studied, we observed absorbance changes appearing within 5 μ s and relaxing exponentially because of charge recombination with a time constant $\tau = 17$ ms (at pH 7.4). From the spectrum of these absorbance changes (Fig. 2a, spectrum A), we conclude that the tryptophanyl radical is in the unprotonated (neutral) state (Trp[•]) at times later than 5 μ s after excitation of FADH[•], because the difference between spectrum A and the (FADH⁻–FADH[•]) difference spectrum B is inconsistent with the absorption spectrum of the protonated (cation) radical TrpH^{•+}, but agrees well with that of Trp[•] (Fig. 2b). The recombination kinetics accelerated strongly upon lowering the pH ($\tau = 7$ ms at pH 6.5 and 0.9 ms at pH 5.4; see Fig. 2a, inset). The most straightforward explanation for the acceleration is that recombination requires Trp[•] reprotonation. The presence of Trp[•] is also consistent with the acidity of TrpH^{•+}, which has a $pK \approx 4$ in water^{14,15}. An electron

transfer from a tyrosine residue to Trp[•] as observed recently in photolyase from *Anacystis nidulans*^{16,17} was not detectable in photolyase from *E. coli*.

To establish whether Trp[•] is formed by hydrogen atom transfer or by electron transfer and subsequent deprotonation, we measured the kinetics of flash-induced absorbance changes in the nanosecond timescale in the 300–645-nm range where deprotonation of TrpH^{•+} is expected to give characteristic absorbance changes (see Fig. 2b). A typical signal measured at 580 nm is shown in the inset of Fig. 3a (trace H₂O). A fast, instrument-limited bleaching is followed by a further bleaching with $\tau \approx 300$ ns. The same kinetic components were observed at all other wavelengths studied. The spectrum of the amplitude of the 300-ns phase (Fig. 3a, spectrum A–B) agrees well

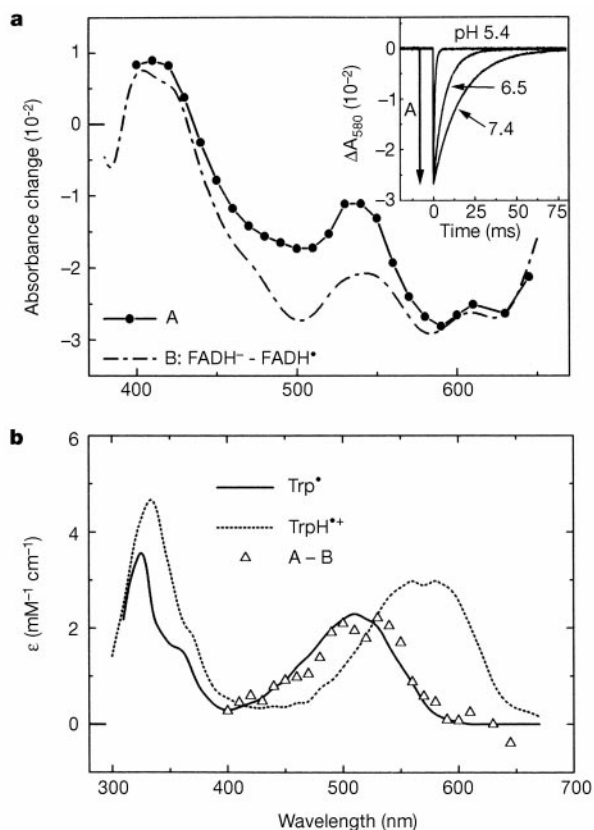


Figure 2 Evidence for a deprotonated tryptophanyl radical (Trp[•]). **a**, Trace A, spectrum of flash-induced absorbance changes decaying with $\tau = 17$ ms (pH 7.4), obtained from kinetic traces like that at pH 7.4 and 580 nm in the inset. Trace B, (FADH⁻–FADH[•]) difference spectrum in *E. coli* photolyase, obtained by subtracting absorption spectra recorded using a Kontron (model UV 922) spectrophotometer before and after saturating illumination with continuous light (540–650 nm) in the presence of 5 mM 1,4-dithiothreitol as exogenous electron donor. Spectrum B was normalized to spectrum A at ~630 nm. Inset, pH effect on the kinetics of the flash-induced absorbance change at 580 nm. The samples contained about 30 μ M photolyase, 200 mM NaCl, 15% (w/w) glycerol and 20 mM buffer (pH 7.4, Tris-HCl; pH 5.4 and pH 6.5, MES). Dithiothreitol was only added in spectrum B. Absorbance changes were recorded with a resolution of 5 μ s using continuous monitoring light¹⁶. The samples were excited by 7-ns dye laser pulses of ~30 mJ cm⁻² at 635 nm (for monitoring wavelengths $\lambda_m \leq 610$ nm) or ~20 mJ cm⁻² at 675 nm (for $\lambda_m \geq 610$ nm). Amplitudes in A were corrected for slight degradation of the samples during data collection and for the lower efficiency of the 675-nm excitation, using control measurements at 580 nm and 610 nm. **b**, Comparison of the difference between A and B (triangles) with absorption spectra of Trp[•] and TrpH^{•+} in aqueous solution (taken from ref. 15). The (A–B) difference spectrum was normalized to the other spectra at their isosbestic point (527 nm). The samples were kept at 10 °C during all spectroscopic measurements.

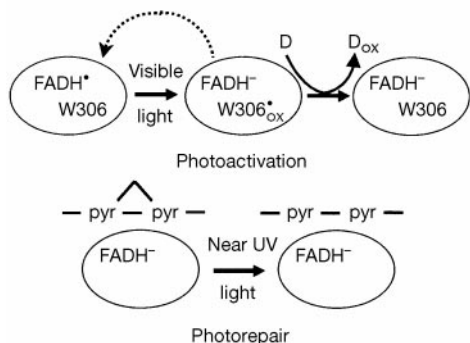


Figure 1 Outline of the photoactivation (top panel) and photorepair (bottom panel) reactions of DNA photolyase. See text for details. D, exogenous electron donor; pyr, pyrimidine in DNA.

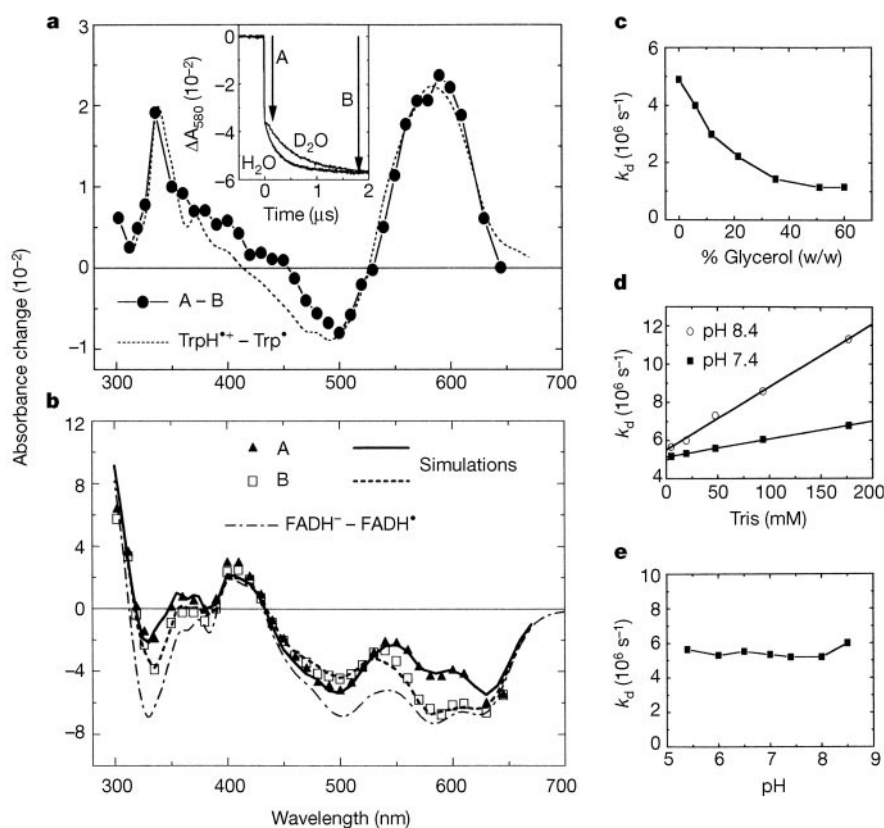


Figure 3 Monitoring the TrpH⁺⁺ deprotonation in the nanosecond time range. **a**, Circles, spectrum of the amplitude of the $\tau \approx 300$ -ns phase obtained from flash-induced absorbance changes like the 'H₂O' trace at 580 nm in the inset. Broken line, (TrpH⁺⁺ - Trp^{*}) difference spectrum calculated from the literature spectra in Fig. 2b and normalized to the spectrum of the 300-ns phase at ~ 580 nm. Inset, Deuterium isotope effect on the kinetics of the absorbance change at 580 nm. The H₂O sample contained 70 μ M photolyase, 200 mM NaCl, 10% (w/w) glycerol and 20 mM Tris, pH 7.4. The D₂O sample was obtained by exchange of H₂O for D₂O (ref. 17) and addition of 10% (v/v) glycerol. Absorbance changes were recorded with a time resolution of 10 ns as described¹⁶. Excitation conditions and correction procedure were as in Fig. 2. **b**, Triangles, spectrum of the absorbance changes preceding the 300-ns phase (amplitude A in the inset of **a**). This spectrum was simulated (solid line) by addition of the TrpH⁺⁺ spectrum (Fig. 2b) and the

(FADH⁻ - FADH^{*}) difference spectrum (dash-dotted line; for details, see Fig. 2a). Squares, spectrum of the absorbance changes after completion of the 300-ns phase, simulated (dashed line) by addition of the Trp^{*} spectrum (Fig. 2b) and the (FADH⁻ - FADH^{*}) difference spectrum. For the best simulations shown, the (FADH⁻ - FADH^{*}) differential molar absorption coefficient at 580 nm was assumed to be -2.4 times the molar absorption coefficient of TrpH⁺⁺ at 560 nm and -3.1 times that of Trp^{*} at 510 nm. **c-e**, Characterization of the deprotonation rate constant k_d of TrpH⁺⁺, obtained by least squares fits to flash-induced absorbance changes at 580 nm. Sample conditions: 30–70 μ M photolyase and 200 mM NaCl; and 20 mM Tris pH 7.4 and glycerol at the concentrations indicated (**c**); Tris pH 7.4 or pH 8.4 at the concentrations indicated (**d**); or 20 mM buffer (pH 5.4, 6 and 6.5, MES; pH 7, 7.5, 8 and 8.5, Tris) (**e**).

with the difference between the solution spectra of TrpH⁺⁺ and Trp^{*}, clearly indicating that the 300-ns phase reflects the deprotonation of TrpH⁺⁺. This conclusion is also supported by the kinetic isotope effect¹⁸ observed upon exchange of H₂O for D₂O: τ (D₂O)/ τ (H₂O) = 2.2 (see Fig. 3a, inset). As a consequence, the 300-ns phase should be preceded by electron transfer from TrpH to excited FADH^{*}, forming TrpH⁺⁺ and FADH⁻. Consistent with this prediction, the spectrum of the absorbance changes preceding the 300-ns phase (Fig. 3b, triangles) can be well simulated (solid line) by the addition of the TrpH⁺⁺ spectrum and the (FADH⁻ - FADH^{*}) difference spectrum.

We conclude that the radical transfer from excited FADH^{*} to the tryptophan residue W306 occurs by fast (< 10 ns) electron transfer (FADH^{*}..TrpH → FADH⁻..TrpH⁺⁺), followed by deprotonation of TrpH⁺⁺ in about 300 ns. The nature of each of these two reactions will now be examined in more detail.

With respect to the deprotonation of the tryptophanyl radical, it is particularly important to identify the proton acceptor. Increasing glycerol concentration in the solvent causes a pronounced decrease in the rate of TrpH⁺⁺ deprotonation (Fig. 3c), suggesting that the proton from TrpH⁺⁺ is released to the solvent rather than being

transferred to a proton accepting residue in the protein. The solvent contained two potential proton acceptors in sufficient concentration to account for a sub-microsecond reaction: unprotonated buffer molecules and H₂O itself. Increasing concentrations of Tris buffer accelerate the deprotonation of TrpH⁺⁺, and this acceleration is much more pronounced at pH 8.4 than at pH 7.4 (Fig. 3d), showing that the unprotonated Tris is the dominating proton acceptor at high concentrations of Tris. Extrapolation to zero Tris concentration yields a deprotonation rate constant, k_d , of about 5.3×10^6 s⁻¹ which we attribute to proton transfer from TrpH⁺⁺ to H₂O in the solvent. Assuming $pK \approx 4$ for the deprotonating TrpH⁺⁺ in photolyase, this rate constant is consistent with the relationship $k_d = k_r \times 10^{-pK}$ s⁻¹ for the deprotonation of weak acids in water, where k_r is the (diffusion controlled) reprotonation rate constant of typically 10^{10} – 10^{11} M⁻¹ s⁻¹ (see ref. 19 for a treatment of proton transfer processes in aqueous media). At the buffer concentration of 20 mM used in all our other experiments, H₂O should still be the dominating proton acceptor (see Fig. 3d). Varying the pH between 5.4 and 8.5 at 20 mM buffer, the kinetics of deprotonation (Fig. 3e) and its amplitude (data not shown) were essentially constant, as expected for H₂O ($pK = -1.7$) as proton acceptor.

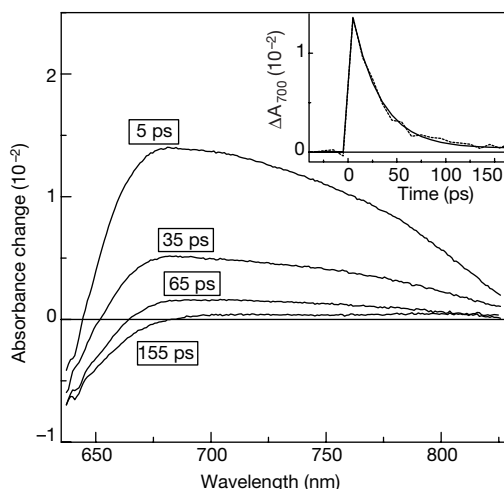


Figure 4 Study of the reaction $\text{FADH}^{\bullet\bullet} \cdot \text{TrpH} \rightarrow \text{FADH}^{\bullet} \cdot \text{TrpH}^{\bullet+}$ by ultrafast spectroscopy. Measurements were performed as described²⁶ with 620-nm pump pulses (70 fs full width at half maximum) and white-light continuum probe pulses, using a continuously moving thermostatted cell with an optical pathlength of 1 mm. Shown are transient spectra at selected delay times. Inset, kinetic data (dashed line) and exponential fit (solid line, $\tau = 29$ ps) at 700 nm. The sample contained 260 μM photolyase, 200 mM NaCl, 15% (w/w) glycerol and 20 mM Tris, pH 7.4.

The kinetics of the electron transfer reaction starting from the excited state $\text{FADH}^{\bullet\bullet}$ were resolved by ultrafast pump-probe spectroscopy in the near-infrared spectral region where an induced absorption has been assigned to $\text{FADH}^{\bullet\bullet}$ (refs. 11, 20). Figure 4 shows typical difference spectra recorded at given delay times, and the kinetic trace at 700 nm. The broad absorbance increase between 650 nm and 800 nm observed immediately after excitation (5-ps spectrum), attributed to $\text{FADH}^{\bullet\bullet}$, decays nearly completely with $\tau \approx 30$ ps (see 700-nm trace). The difference spectrum observed after completion of the 30-ps phase (see 155-ps spectrum) consists of a bleaching below 680 nm (the ground state absorption of

$\text{FADH}^{\bullet}$) and is consistent with the presence of the state $(\text{FADH}^{\bullet} \cdot \text{TrpH}^{\bullet+})$ (see Fig. 3b). These results show that the excited state $\text{FADH}^{\bullet\bullet}$ decays in about 30 ps, presumably by abstracting an electron from a nearby tryptophan residue (see below).

Figure 5 depicts a reaction scheme for the radical transfer from excited $\text{FADH}^{\bullet}$ to W306 that is based on our results and takes into account essential elements of the X-ray structure of *E. coli* photolyase²¹. The solvent accessible tryptophan residue W306 is as far as 13.4 Å (shortest edge-to-edge distance) from the flavin, excluding a direct or superexchange-mediated electron transfer to $\text{FADH}^{\bullet\bullet}$ in 30 ps (refs. 7, 22). However, two other tryptophan residues, W382 and W359, are located in between (Fig. 5, inset), suggesting an electron transfer in three steps. The first step, downhill electron transfer from W382 to $\text{FADH}^{\bullet\bullet}$ takes 30 ps, the observed lifetime of $\text{FADH}^{\bullet\bullet}$ (Fig. 4). The next two steps, the presumably nearly isoenergetic electron transfers from W359 to the cation radical of W382 and from W306 to the cation radical of W359, should be faster than 10 ns because otherwise the deprotonation of the cation radical of W306, measured with a time resolution of 10 ns (Fig. 3a), should have shown a lag phase. Given the short distances between the partners (Fig. 5, inset), these three electron transfer rates are consistent with established rates of other intraprotein electron transfer reactions⁷. The proton release from the cation radical of W306 to the aqueous medium in about 300 ns is driven by a substantial decrease in free energy (about 0.2 eV at pH 7.4, assuming a pK of 4 for $\text{TrpH}^{\bullet+}$). This energetic stabilization is essential to trap the oxidizing radical on W306, which is accessible to exogenous reductants and sufficiently far apart from $\text{FADH}^{\bullet}$ to prevent a fast direct recombination; moreover, the deprotonation creates a free energy barrier in a potential recombination pathway through repopulation of the cation radical of W306 and reverse electron transfer in the chain $\text{FAD}^{\bullet\bullet}$ –W382–W359–W306 (Fig. 5, broken arrows). The acceleration of the recombination with decreasing pH (Fig. 2a, inset) indicates that this is the actual recombination pathway.

Examination of the X-ray structure of *E. coli* photolyase²¹ reveals important differences between the environments of the three tryptophans. The indole ring of W306 is partially surrounded by mainly polar amino acids. Its ring nitrogen is directly exposed to a shallow pocket that is open to the solvent and that can accommodate

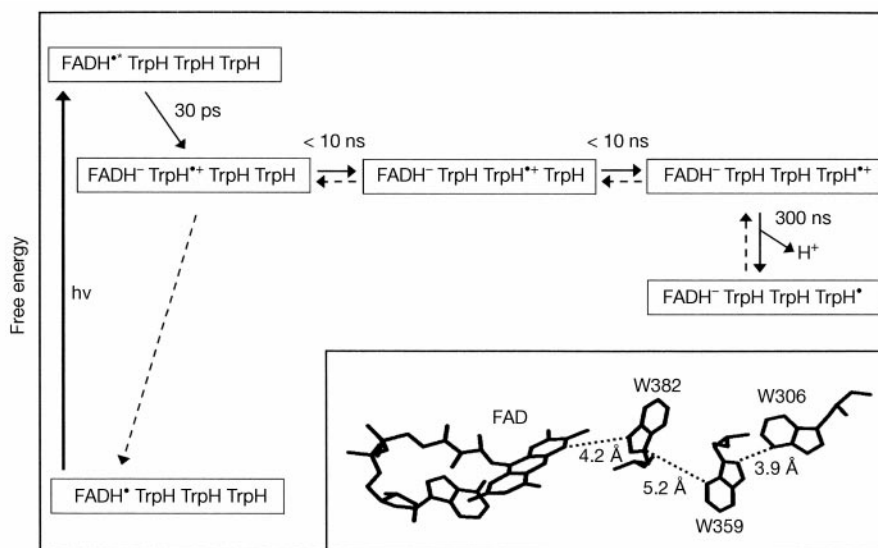


Figure 5 Reaction scheme for the radical transfer during photoactivation of *E. coli* photolyase. TrpH TrpH TrpH refers (from left to right) to W382, W359 and W306. Productive (forward) reactions are indicated as solid arrows, and back reactions as broken arrows. The free energy levels are not drawn to scale. We estimate the excited state of $\text{FADH}^{\bullet}$ to be ~ 2.0 eV above the ground state (corresponding to the energy of a 630-nm

photon), and the $(\text{FADH}^{\bullet} \cdot \text{TrpH}^{\bullet+})$ states to be ~ 1.5 eV (assuming midpoint potentials of about -0.4 V for $\text{FADH}^{\bullet\bullet}/\text{FADH}^{\bullet}$ in photolyase²⁷ and 1.1 V for $\text{TrpH}^{\bullet+}/\text{TrpH}$ buried in a protein¹⁴). Inset, arrangement and shortest edge-to-edge distances between FAD, W382, W359 and W306 obtained from the X-ray structure of photolyase from *E. coli*²¹. Swiss-PdbViewer²⁸ was used to prepare the inset.

a water molecule, a situation that should favour proton transfer from the W306 cation radical to the solvent. In contrast, W382 is buried in the hydrophobic interior of the protein. There is no potential proton acceptor within 5 Å of the ring nitrogen of W382, excluding the possibility of deprotonation of the cation radical. The indole ring of W359 is surrounded by both hydrophobic and polar amino acids. The ring nitrogen appears to be hydrogen bonded to a more buried water molecule that might assist deprotonation of the W359 cation radical; however, our experimental data with a time resolution of 10 ns, together with previous data on the W306F mutant¹², do not give any indication for a deprotonation of the W359 cation radical.

The different polarities of the tryptophan environments should be relevant for the energetics of the electron transfer, as the reduction potential of the TrpH⁺/TrpH couple is higher in a hydrophobic protein environment than in water¹⁴. For this reason, there is likely to be some decrease of the free energy of the TrpH⁺ states in the order W382, W359, W306, favouring radical localization on W306 already, before its deprotonation.

A reaction scheme quite different from Fig. 5 was proposed previously⁹. It assumed that a long-lived quartet state of FADH[•] (formed by intersystem crossing from the initial excited doublet state) abstracts an electron from W306 in about 500 ns, and that the resulting cation radical TrpH⁺ remains protonated. This mechanism is incompatible with our results which establish unambiguously that TrpH⁺ is formed in less than 10 ns and deprotonates in about 300 ns. The origin of the disagreement is most probably that the previous scheme was based on studies^{11,20,23} using 355-nm laser flashes or Xe flashes that could excite species other than FADH[•] (for example, fully oxidized FAD; see ref. 24). Under our excitation conditions, FADH[•] is the only species being excited.

In summary, we have reached a consistent mechanism for the radical transfer during photoactivation of photolase in which sequential electron transfer among amino-acid residues generates transient cation radicals before charge neutralization by proton release to the solvent. Thus, we conclude that charge compensating simultaneous proton transfer is not a prerequisite for intraprotein radical transfer. A definitive answer to the question of whether long-range radical transfer in other enzymes proceeds through a similar mechanism may require time-resolved methods to selectively monitor electron transfer, H-atom transfer and protonation/deprotonation events.

Our results on the activation of DNA photolase from *E. coli* may be relevant for the other photolases (class I and class II)¹⁰ and also for the cryptochrome blue-light receptors in plants and animals, which are flavoproteins similar in sequence to photolases²⁵. Indeed, inspecting the aligned amino-acid sequences¹⁰ of known photolases and cryptochromes, we noticed that the tryptophan residues W382, W359 and W306 are uniformly conserved. This observation suggests that the electron transfer chain described here is involved in the function of all photolases, and of blue light receptors as well. □

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- Stubbe, J. & van der Donk, W. A. Protein radicals in enzyme catalysis. *Chem. Rev.* **98**, 705–762 (1998).
- Uhlir, U. & Eklund, H. Structure of ribonucleotide reductase protein R1. *Nature* **370**, 533–539 (1994).
- Sjöberg, B.-M. The ribonucleotide reductase jigsaw puzzle: a large piece falls into place. *Structure* **2**, 793–796 (1994).
- Siegbahn, P. E. M., Blomberg, M. R. A. & Crabtree, R. H. Hydrogen transfer in the presence of amino acid radicals. *Theor. Chem. Acc.* **97**, 289–300 (1997).
- Siegbahn, P. E. M., Eriksson, L., Himo, F. & Pavlov, M. Hydrogen atom transfer in ribonucleotide reductase (RNR). *J. Phys. Chem.* **102**, 10622–10629 (1998).
- Cukier, R. I. & Nocera, D. G. Proton coupled electron transfer. *Annu. Rev. Phys. Chem.* **49**, 337–369 (1998).
- Page, C. C., Moser, C. C., Chen, X. & Dutton, P. L. Natural engineering principles of electron tunnelling in biological oxidation-reduction. *Nature* **402**, 47–52 (1999).
- Sancar, A. No “end of history” for photolases. *Science* **272**, 48–49 (1996).
- Kim, S. -T., Heelis, P. F. & Sancar, A. Role of tryptophans in substrate binding and catalysis by DNA photolase. *Methods Enzymol.* **258**, 319–343 (1995).

- Yasui, A. & Eker, A. P. M. in *DNA Damage and Repair*, Vol. 2: *DNA Repair in Higher Eukaryotes* (eds Nickoloff, J. A. & Hoekstra, M. E.) 9–32 (Humana, Totowa, 1998).
- Heelis, P. F., Okamura, T. & Sancar, A. Excited-state properties of *Escherichia coli* DNA photolase in the picosecond to millisecond time scale. *Biochemistry* **29**, 5694–5698 (1990).
- Li, Y. F., Heelis, P. F. & Sancar, A. Active site of DNA photolase: tryptophan-306 is the intrinsic hydrogen atom donor essential for flavin radical photoreduction and DNA repair *in vitro*. *Biochemistry* **30**, 6322–6329 (1991).
- Eker, A. P. M., Yajima, H. & Yasui, A. DNA photolase from the fungus *Neurospora crassa*. Purification, characterization and comparison with other photolases. *Photochem. Photobiol.* **60**, 125–133 (1994).
- Tommos, C., Skaliky, J. J., Pilloud, D. L., Wand, A. J. & Dutton, P. L. *De novo* proteins as models of radical enzymes. *Biochemistry* **38**, 9495–9507 (1999).
- Solar, S., Getoff, G., Surdhar, P. S., Armstrong, D. A. & Singh, A. Oxidation of tryptophan and N-methylindole by N₃⁺, Br₂⁺, and (SCN)₂⁺ radicals in light- and heavy-water solutions: A pulse radiolysis study. *J. Phys. Chem.* **95**, 3636–3643 (1991).
- Aubert, C., Mathis, P., Eker, A. P. M. & Brettel, K. Intraprotein electron transfer between tyrosine and tryptophan in DNA photolase from *Anacystis nidulans*. *Proc. Natl Acad. Sci. USA* **96**, 5423–5427 (1999).
- Aubert, C., Brettel, K., Mathis, P., Eker, A. P. M. & Boussac, A. EPR detection of the transient tyrosyl radical in DNA photolase from *Anacystis nidulans*. *J. Am. Chem. Soc.* **121**, 8659–8660 (1999).
- Cleland, W. W., O’Leary, M. H. & Northrop, D. B. *Isotope Effects on Enzyme-Catalyzed Reactions* (Univ. Park Press, Baltimore, London, Tokyo, 1977).
- Eigen, M. Proton transfer, acid-base catalysis, and enzymatic hydrolysis Part I: elementary processes. *Angew. Chem. Internat. Edn.* **3**, 1–19 (1964).
- Okamura, T., Sancar, A., Heelis, P. F., Hirata, Y. & Mataga, N. Doublet-quartet intersystem crossing of flavin radical in DNA photolase. *J. Am. Chem. Soc.* **111**, 5967–5969 (1989).
- Park, H. W., Kim, S. T., Sancar, A. & Deisenhofer, J. Crystal structure of DNA photolase from *Escherichia coli*. *Science* **268**, 1866–1872 (1995).
- Cheung, M. S., Daizadeh, I., Stuchebrukhov, A. A. & Heelis, P. F. Pathways of electron transfer in *Escherichia coli* DNA photolase: Trp³⁰⁶ to FADH. *Biophys. J.* **76**, 1241–1249 (1999).
- Kim, S. T., Sancar, A., Essenmacher, C. & Babcock, G. T. Time-resolved EPR studies with DNA photolase: excited-state FADH[•] abstracts an electron from Trp-306 to generate FADH[•], the catalytically active form of the cofactor. *Proc. Natl Acad. Sci. USA* **90**, 8023–8027 (1993).
- Gindt, Y. M. *et al.* Origin of the transient electron paramagnetic resonance signals in DNA photolase. *Biochemistry* **38**, 3857–3866 (1999).
- Cashmore, A. R., Jarillo, J. A., Xu, Y. -J. & Liu, D. Cryptochromes: blue light receptors for plants and animals. *Science* **284**, 760–765 (1999).
- Martin, J.-L. & Vos, M. H. Femtosecond measurements of geminate recombination in heme proteins. *Methods Enzymol.* **232**, 416–430 (1994).
- Heelis, P. F., Deeble, D. J., Kim, S. T. & Sancar, A. Splitting of cys-syn cyclobutane thymine-thymine dimers by radiolysis and its relevance to enzymatic photoreactivation. *Int. J. Radiat. Biol.* **62**, 137–143 (1992).
- Guex, N. & Peitsch, M. N. SWISS-MODEL and the Swiss-PdbViewer: An environment for comparative protein modeling. *Electrophoresis* **18**, 2714–2723 (1997).

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Engineering stability in gene networks by autoregulation

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The genetic and biochemical networks which underlie such things as homeostasis in metabolism and the developmental programs of living cells, must withstand considerable variations and random perturbations of biochemical parameters^{1–3}. These occur as transient changes in, for example, transcription, translation, and RNA and protein degradation. The intensity and duration of these perturbations differ between cells in a population⁴. The unique state of cells, and thus the diversity in a population, is owing to the different environmental stimuli the individual cells experience and the inherent stochastic nature of biochemical processes (for example, refs 5 and 6). It has been proposed, but not

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